


Engineering Electromagnetics

工程电磁场

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8: Energy and Potential 能量与电位

- Conservative Field 保守场

A field is conservative if its line integral between any two points is independent of the path chosen. Another property of a conservative field is that its *closed path* line integral is zero: 场量的线积分与积分路径无关，或者场量的闭合路径积分为零：

$$\oint \mathbf{F} \cdot d\mathbf{L} = 0$$

Note: Conservative field must be vector field! 保守场一定是矢量场！

Most fields in nature are conservative, such as earth's gravitational field, electrostatic field...

One can confirm that a field is conservative by evaluating every possible closed path integral and showing a result of zero in all cases. A much faster way, that we will see later, is to evaluate the *curl* 旋度 of the field. If the curl is zero at all defined points, the field is conservative.

Note: Conservative field must be irrotational field! 保守场一定是无旋场！

- Conservative Force 保守力

$$\oint \mathbf{F} \cdot d\mathbf{L} = 0$$

A conservative force is a force with the property that the work done in moving a particle between two points is independent of the taken path. Equivalently, if a particle travels in a closed loop, the net work done by a conservative force is zero.

Note: In other words, the force involved in conservative field is conservative force, and sometimes, conservative field is also termed as ‘conservative force field’.

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- Potential Field of a Point Charge 点电荷的电位场

We just found the difference in potential between two positions in a point charge field:

$$V_{AB} = V_A - V_B = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right)$$

We could perform the same calculation by specifying the starting point at infinity, and the ending point at some general radius, r:

$$\boxed{V_{r\infty} = V_r - V_\infty = -\int_\infty^r \mathbf{E}\cdot d\mathbf{L} = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{r} - \frac{1}{\infty}\right) = \frac{Q}{4\pi\epsilon_0 r}}$$

This result is a *potential function* or *potential field*, which specifies potential at any position within the defined space, and with the potential at infinity (the reference value) equal to zero.

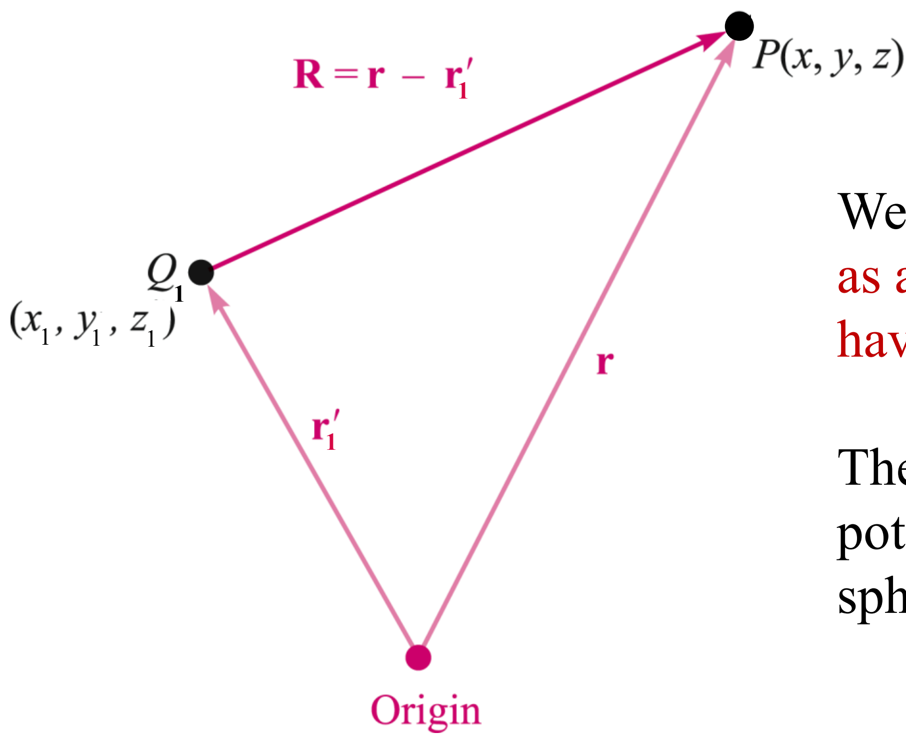
In practice, we can "bias" this function any way we want (or need) to, by an additive constant, C_1:

$$V = \frac{Q}{4\pi\epsilon_0 r} + C_1$$

- Potential Field of a Point Charge Off-Origin

The setup here is the same as what we used in writing the electric field of an off-origin point charge.

$$V_P(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}_1|}$$



We now define an *equipotential surface* as a surface composed of all those points having the same value of potential.

The equipotential surfaces in the potential field of a point charge are spheres centered at the point charge.

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- Potential Field Arising From Two or More Point Charges

Introduce a second point charge, and the two scalar potentials simply add:

$$V(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_1|} + \frac{Q_2}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_2|}$$

For n charges, the process continues:

$$\boxed{V(\mathbf{r}) = \sum_{m=1}^{n} \frac{Q_m}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_m|}}$$

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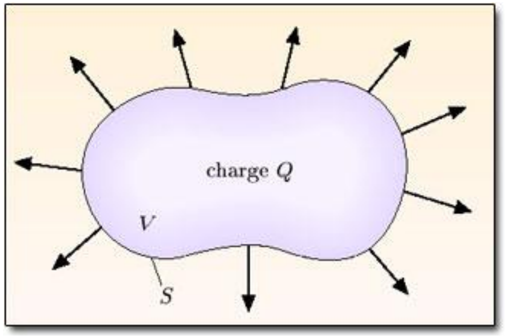
- Potential Associated with Continuous Charge Distributions

If each point charge is now represented as a small element of a continuous volume charge distribution $\rho_\nu \Delta \nu$, then



$$V(\mathbf{r}) = \frac{\rho_\nu(\mathbf{r}_1)\Delta \nu_1}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}_1|} + \frac{\rho_\nu(\mathbf{r}_2)\Delta \nu_2}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}_2|} + \cdots + \frac{\rho_\nu(\mathbf{r}_n)\Delta \nu_n}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}_n|}$$

As we allow the number of elements to become infinite, we obtain the integral expression:

$$\boxed{V(\mathbf{r}) = \int_{\mathrm{vol}} \frac{\rho_\nu(\mathbf{r}')\,d\nu'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}} \qquad \boxed{\mathbf{E}(\mathbf{r}) = \int_{\mathrm{vol}} \frac{\rho_\nu(\mathbf{r}')\,d\nu'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|^2}\frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|}}$$

$$\boxed{\mathbf{D} = \epsilon_0 \mathbf{E}} \quad \text{(free space only)}$$

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- Potential Functions Associated with Line, Surface, and Volume Charge Distributions

Line Charge: $$V(\mathbf{r}) = \int \frac{\rho_L(\mathbf{r}')\,dL'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}$$

Surface Charge: $$V(\mathbf{r}) = \int_S \frac{\rho_S(\mathbf{r}')\,dS'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}$$

Volume Charge: $$V(\mathbf{r}) = \int_{\text{vol}} \frac{\rho_v(\mathbf{r}')\,dv'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}$$

- Example

The problem is to find the potential anywhere on the z axis arising from a circular ring of charge in the x-y plane, centered at the origin.

We use:

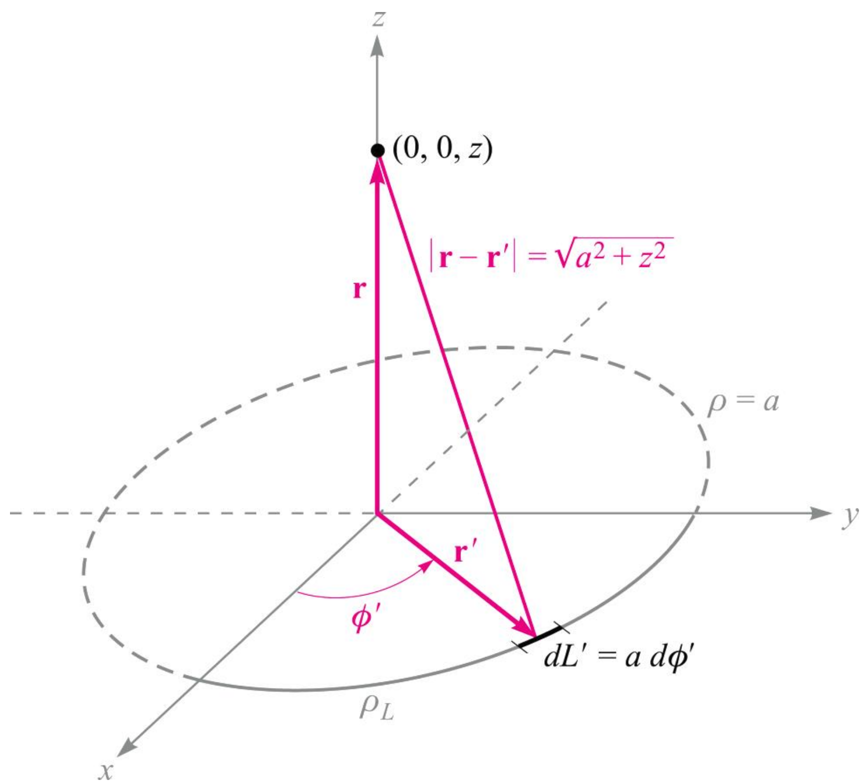
$$V(\mathbf{r}) = \int \frac{\rho_L(\mathbf{r}')\,dL'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}$$

with $dL' = ad\phi'$

$$\mathbf{r} = z\mathbf{a}_z$$

$$\mathbf{r}' = a\mathbf{a}_\rho$$

$$|\mathbf{r}-\mathbf{r}'| = \sqrt{a^2+z^2}$$



- Example (Cont'd)

So now $V(\mathbf{r}) = \int \frac{\rho_L(\mathbf{r}')\,dL'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|}$

becomes:

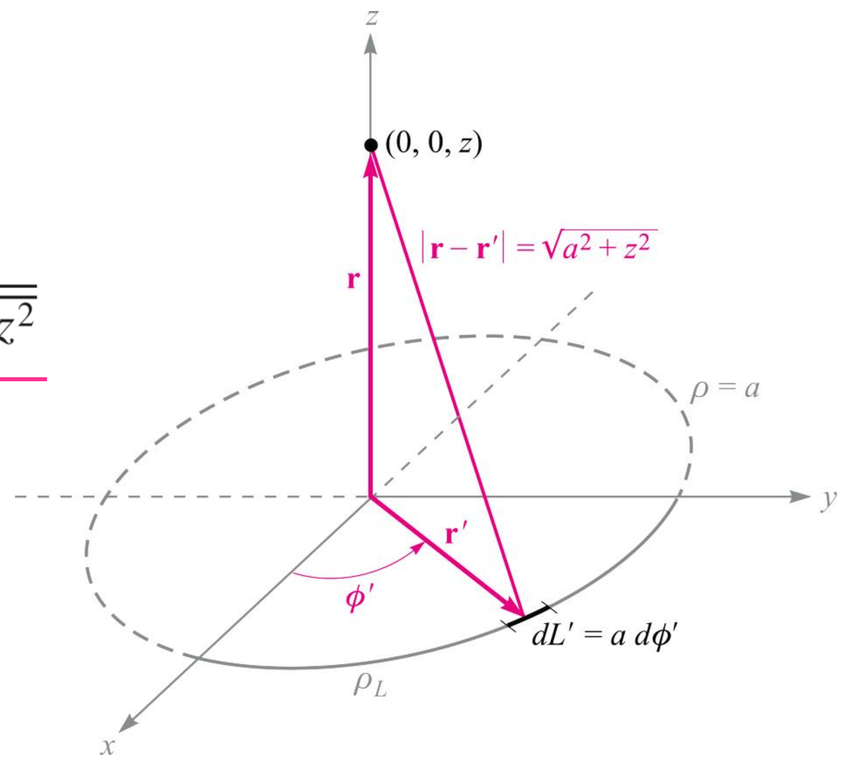
$$V = \int_0^{2\pi} \frac{\rho_L a\,d\phi'}{4\pi\epsilon_0\sqrt{a^2+z^2}} = \frac{\rho_L a}{2\epsilon_0\sqrt{a^2+z^2}}$$

where $dL' = ad\phi'$

$\mathbf{r} = z\mathbf{a}_z$

$\mathbf{r}' = a\mathbf{a}_\rho$

$|\mathbf{r}-\mathbf{r}'| = \sqrt{a^2+z^2}$



- Summary

For a zero reference at infinity, then:

1. The potential arising from a single point charge is the work done in carrying a unit positive charge from infinity to the point at which we desire the potential, and the work is independent of the path chosen between those two points.
2. The potential field in the presence of a number of point charges is the sum of the individual potential fields arising from each charge.
3. The potential arising from a number of point charges or any continuous charge distribution may therefore be found by carrying a unit charge from infinity to the point in question along any path we choose.

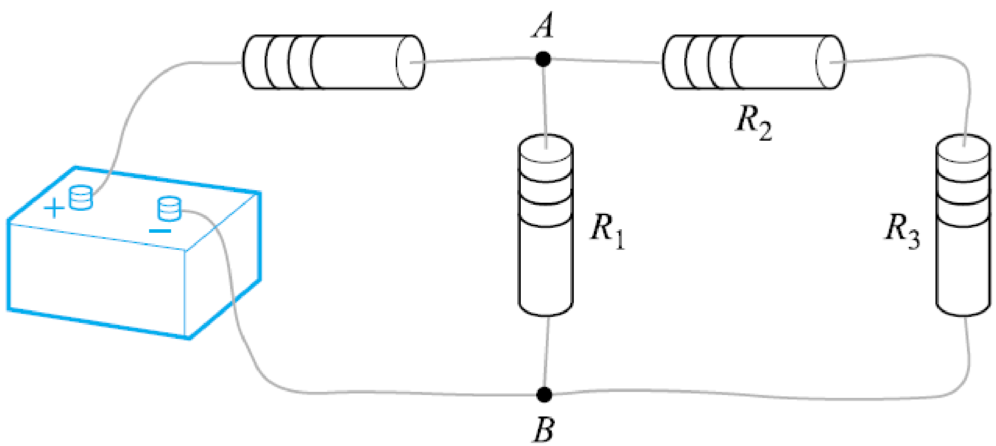
In other words, the expression for potential (zero reference at infinity),

$$V_A = -\int_{\infty}^{A} \mathbf{E} \cdot d\mathbf{L}$$

$$V_{AB} = V_A - V_B = -\int_{B}^{A} \mathbf{E} \cdot d\mathbf{L} \Rightarrow \boxed{\oint \mathbf{E} \cdot d\mathbf{L} = 0}$$

- Summary

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

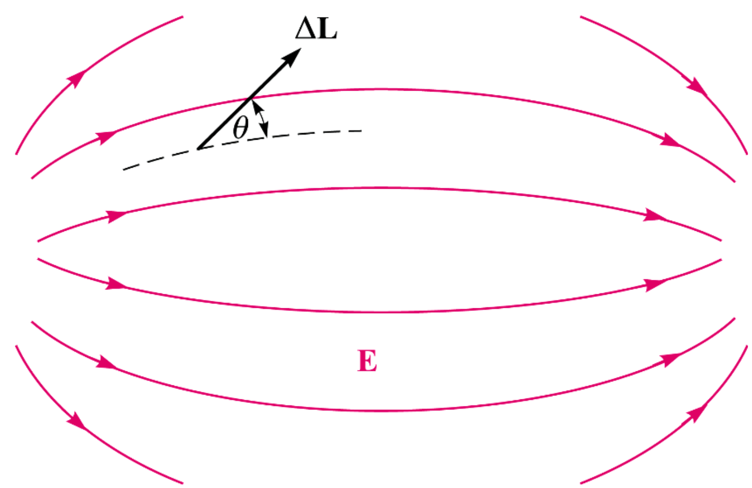


Kirchhoff's circuital law for voltages

- Change in Potential over an Incremental Distance

$$V(\mathbf{r}) = \int_{\mathrm{vol}} \frac{\rho_v(\mathbf{r}')\,dv'}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}'|} \qquad V_A = -\int_{\infty}^{A} \mathbf{E}\cdot d\mathbf{L}$$

The change in potential occurring over distance $\Delta\mathbf{L}$ depends on the angle between this vector and the electric field; i.e., the projection of the field along the path:



$$\Delta V \doteq -\mathbf{E}\cdot\Delta\mathbf{L}$$

or $\Delta V \doteq -E\,\Delta L\cos\theta$

from which: $\dfrac{dV}{dL} = -E\cos\theta$

whose maximum value is: $\left.\dfrac{dV}{dL}\right|_{\max} = E$

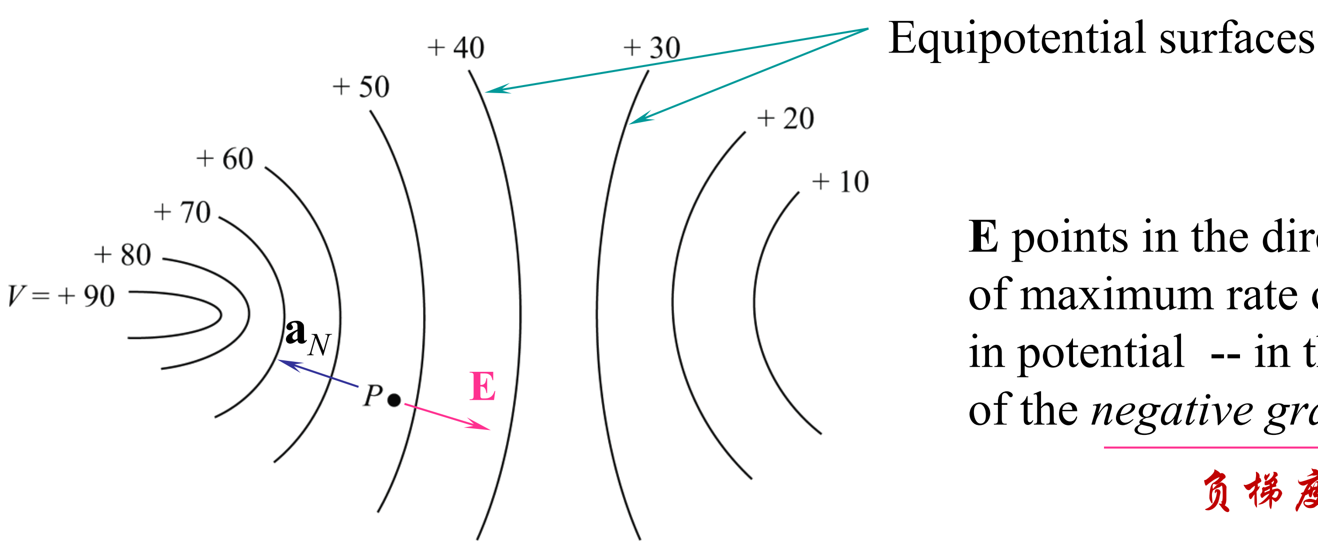
when the path vector lies *along* the electric field direction.

- Relation Between Potential and Electric Field

The maximum rate of *increase* in potential should occur in a direction *exactly opposite* the electric field:

$$\mathbf{E} = -\frac{dV}{dL}\bigg|_{\max} \mathbf{a}_N$$

$\mathbf{a}_N$: unit vector normal to an equipotential surface and in the direction of *increasing* potential



E points in the direction of maximum rate of *decrease* in potential -- in the direction of the *negative gradient* of *V*.

负梯度

- Electric Field in Terms of *V* in Rectangular Coordinates

$$\mathbf{E} = -\left.\frac{dV}{dL}\right|_{\max} \mathbf{a}_N \qquad dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$$

We also know that: $dV = -\mathbf{E}\cdot d\mathbf{L} = -E_x\,dx - E_y\,dy - E_z\,dz$

We therefore identify:

$$\begin{cases} E_x = -\dfrac{\partial V}{\partial x} \\ E_y = -\dfrac{\partial V}{\partial y} \\ E_z = -\dfrac{\partial V}{\partial z} \end{cases}$$

So that:

$$\boxed{\mathbf{E} = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right)}$$

• Electric Field in Terms of V in Rectangular Coordinates

$$\boxed{\mathbf{E} = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right)}$$

This is obtained by using the del operator, $\nabla = \frac{\partial}{\partial x}\mathbf{a}_x + \frac{\partial}{\partial y}\mathbf{a}_y + \frac{\partial}{\partial z}\mathbf{a}_z$ on V

A more compact relation therefore emerges, which is applicable to *static* electric fields:

$$\boxed{\mathbf{E} = -\nabla V}$$

E is equal to the *negative gradient* of V

$$\boxed{\mathbf{E} = -\text{grad } V}$$

The direction of the gradient is that of the maximum rate of *increase* in the scalar field, or normal to all equipotential surfaces.

Note: 1) The gradient of a scalar is a vector!

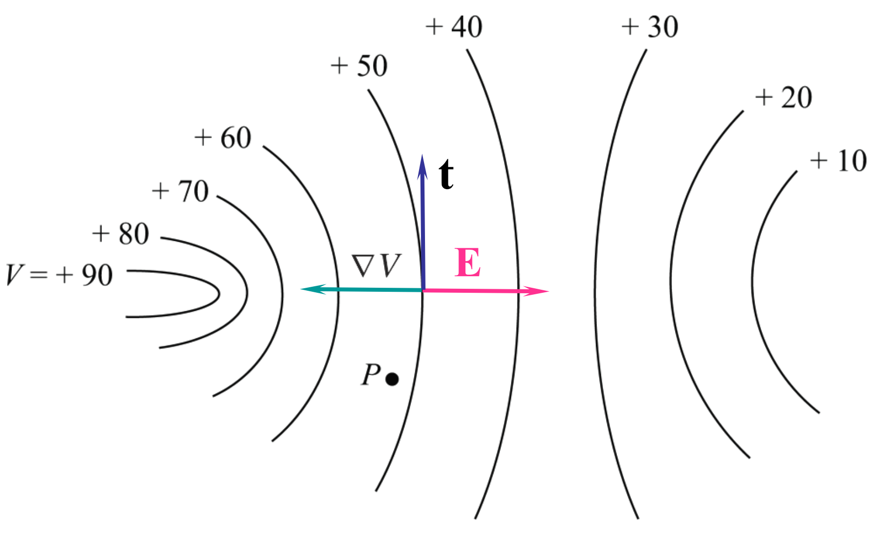
2) Any gradient field is conservative field, or irrotational field.

• Direction of the Gradient Vector

The projection of the gradient along a direction tangent to an equipotential surface *must* give a result of zero, as the potential by definition is constant along that surface: In other words,

$$\nabla V \cdot \mathbf{t} = 0$$

Therefore, ∇V must be perpendicular to **t**, or *normal* to an equipotential surface, and in the direction of *maximum increase* in *V.*



- Gradient of V in the Three Coordinate Systems

$$\nabla V = \frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z \qquad \text{(rectangular)}$$

$$\nabla V = \frac{\partial V}{\partial \rho}\mathbf{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi + \frac{\partial V}{\partial z}\mathbf{a}_z \qquad \text{(cylindrical)}$$

$$\nabla V = \frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi \qquad \text{(spherical)}$$

- Example

Given the potential field, $V = 2x^2y - 5z$, and a point $P(-4, 3, 6)$, we wish to find several numerical values at point P: the potential V, the electric field intensity $\mathbf{E}$, the direction of $\mathbf{E}$, the electric flux density $\mathbf{D}$, and the volume charge density ρ_v.

- Example

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Solution. The potential at $P(-4, 5, 6)$ is

$$V_P = 2(-4)^2(3) - 5(6) = 66 \text{ V}$$

Next, we may use the gradient operation to obtain the electric field intensity,

$$\mathbf{E} = -\nabla V = -4xy\mathbf{a}_x - 2x^2\mathbf{a}_y + 5\mathbf{a}_z \text{ V/m}$$

The value of $\mathbf{E}$ at point P is

$$\mathbf{E}_P = 48\mathbf{a}_x - 32\mathbf{a}_y + 5\mathbf{a}_z \text{ V/m}$$

and

$$|\mathbf{E}_P| = \sqrt{48^2 + (-32)^2 + 5^2} = 57.9 \text{ V/m}$$

The direction of $\mathbf{E}$ at P is given by the unit vector

$$\begin{aligned}\mathbf{a}_{E,P} &= (48\mathbf{a}_x - 32\mathbf{a}_y + 5\mathbf{a}_z)/57.9 \\ &= 0.829\mathbf{a}_x - 0.553\mathbf{a}_y + 0.086\mathbf{a}_z\end{aligned}$$

- ## Example (Cont’d)

Given the potential field, $V = 2x^2y - 5z$, and a point $P(-4, 3, 6)$, we wish to find several numerical values at point P: the potential V, the electric field intensity $\mathbf{E}$, the direction of $\mathbf{E}$, the electric flux density $\mathbf{D}$, and the volume charge density ρ_v.

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If we assume these fields exist in free space, then

$$\mathbf{D} = \epsilon_0\mathbf{E} = -35.4xy\,\mathbf{a}_x - 17.71x^2\,\mathbf{a}_y + 44.3\,\mathbf{a}_z \text{ pC/m}^{\cancel{3}\,2}$$

Finally, we may use the divergence relationship to find the volume charge density that is the source of the given potential field,

$$\rho_v = \nabla\cdot\mathbf{D} = -35.4y \text{ pC/m}^3$$

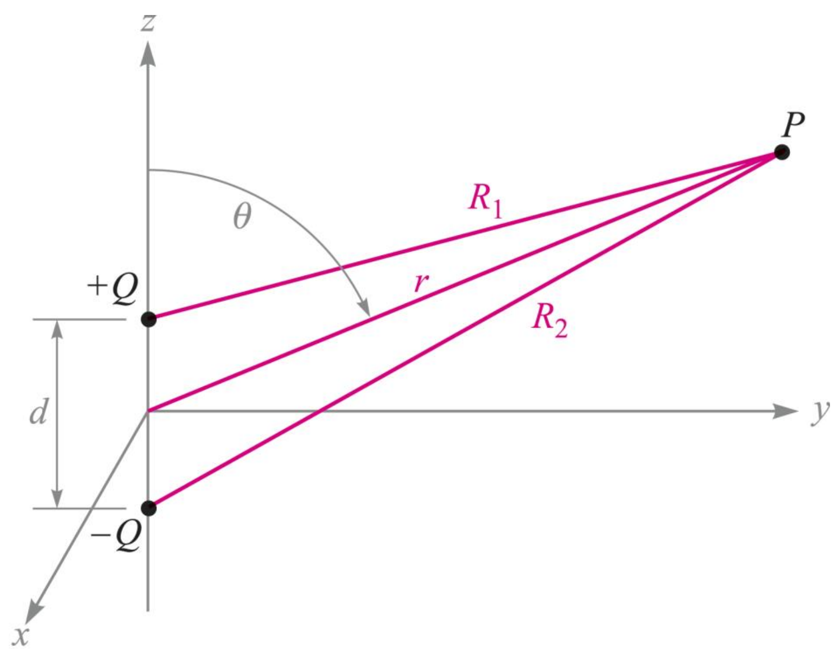
At P, $\rho_v = -106.2 \text{ pC/m}^3$.

- Electric Dipole 电偶极子

An *electric dipole*, or simply a *dipole*, is the name given to two point charges of equal magnitude and opposite sign, separated by a distance that is small compared to the distance to the point P at which we want to know the electric and potential fields.

The potential will be just the sum of the two potential functions associated with each point charge:

$$V = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{Q}{4\pi\epsilon_0}\frac{R_2 - R_1}{R_1 R_2}$$



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- Far-Field Approximation 远场近似

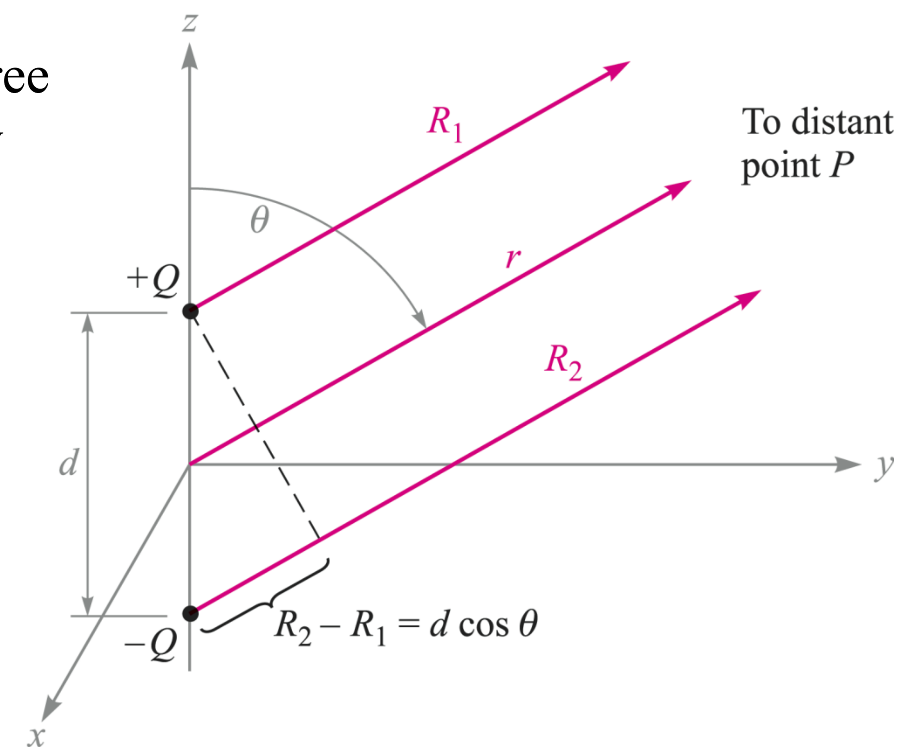
Under the condition $r >> d$, the three position vectors are approximately parallel.

This means that we may use the approximations:

$$R_1 R_2 \doteq r^2$$

and $R_2 - R_1 \doteq d \cos\theta$

to get finally:



$$\boxed{V = \frac{Qd\cos\theta}{4\pi\epsilon_0 r^2}}$$

$$V = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{Q}{4\pi\epsilon_0}\frac{R_2 - R_1}{R_1 R_2}$$

- Electric Field of the Dipole

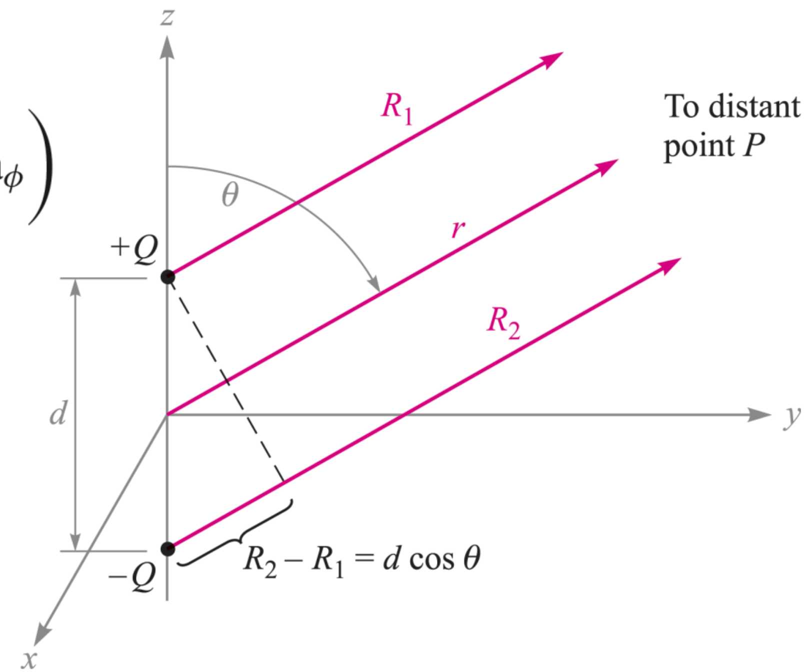
$$\boxed{V = \frac{Qd\cos\theta}{4\pi\epsilon_0 r^2}}$$

$$\mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi\right)$$

or.. $$\mathbf{E} = -\left(-\frac{Qd\cos\theta}{2\pi\epsilon_0 r^3}\mathbf{a}_r - \frac{Qd\sin\theta}{4\pi\epsilon_0 r^3}\mathbf{a}_\theta\right)$$

from which finally:

$$\boxed{\mathbf{E} = \frac{Qd}{4\pi\epsilon_0 r^3}(2\cos\theta\,\mathbf{a}_r + \sin\theta\,\mathbf{a}_\theta)}$$



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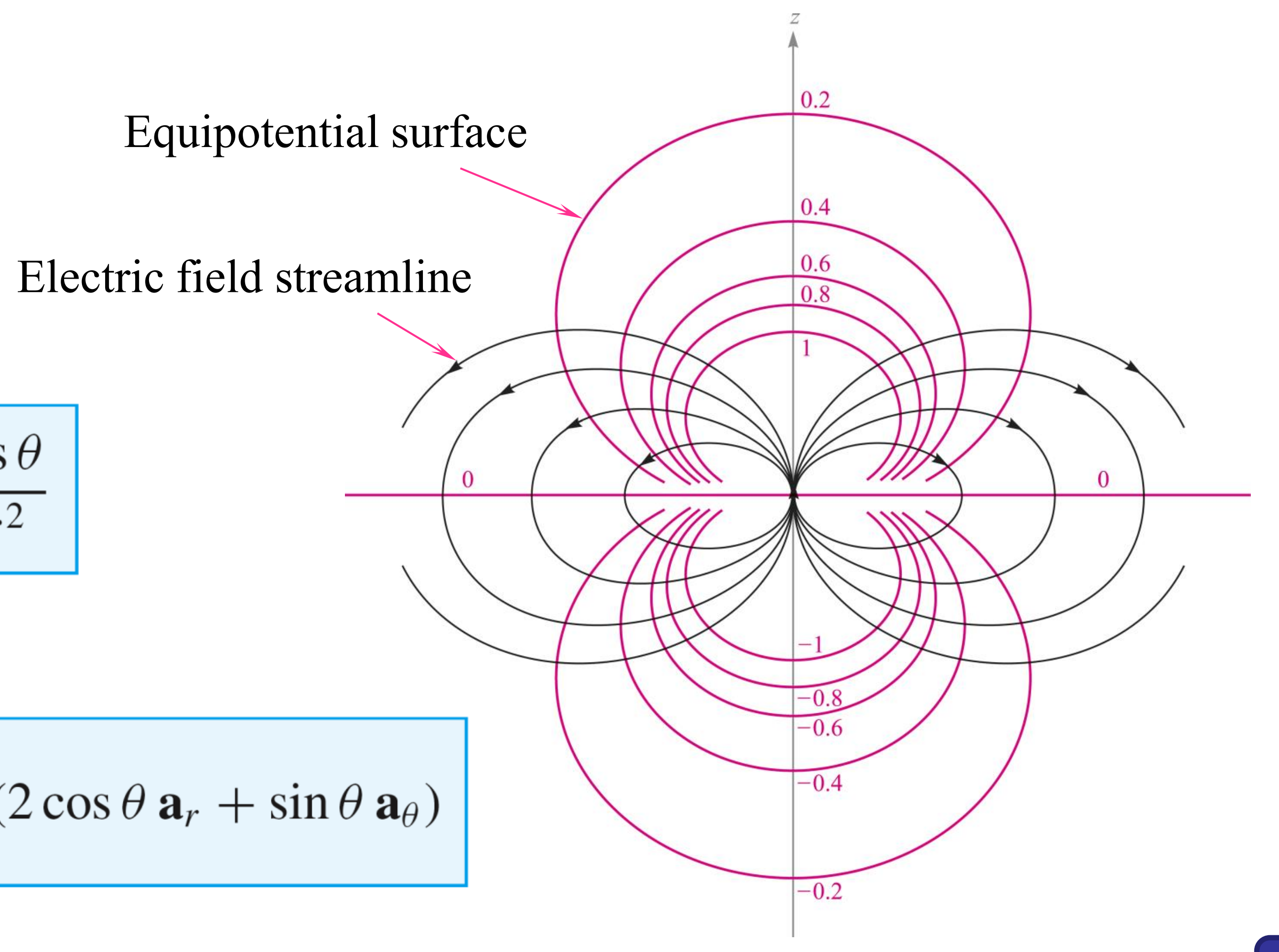
- Electric Dipole Field and Equipotentials



$$V = \frac{Qd\cos\theta}{4\pi\epsilon_0 r^2}$$

$$\mathbf{E} = \frac{Qd}{4\pi\epsilon_0 r^3}(2\cos\theta\,\mathbf{a}_r + \sin\theta\,\mathbf{a}_\theta)$$

- Electric Dipole Moment 偶极矩

The dipole moment vector is directed from the negative charge to the positive charge, and is defined as:

$$\boxed{\mathbf{p} = Q\mathbf{d}}$$ The units of $\mathbf{p}$ are C · m.

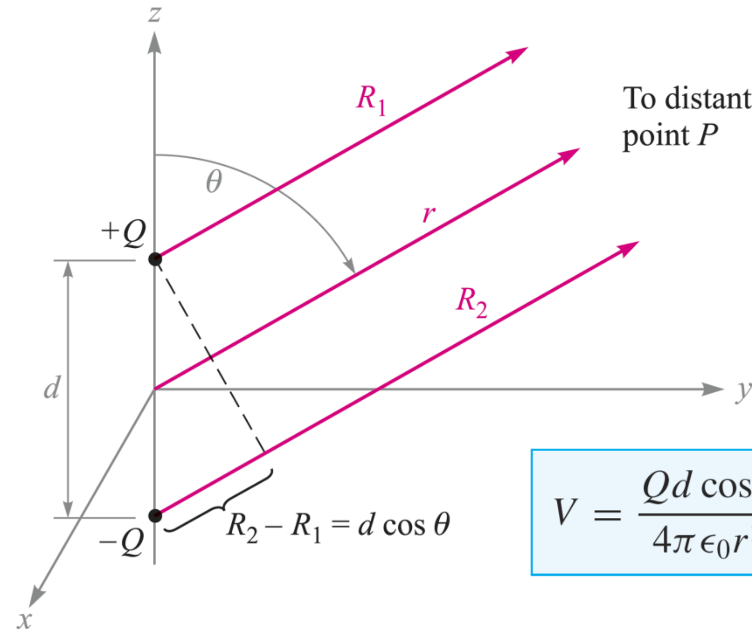
In the charge configuration we have used, the direction of $\mathbf{p}$ is $\mathbf{a}_z$, and therefore:

$$\mathbf{d} \cdot \mathbf{a}_r = d \cos\theta$$

so we may write:

$$V = \frac{\mathbf{p} \cdot \mathbf{a}_r}{4\pi\epsilon_0 r^2}$$

which would account for any orientation of $\mathbf{p}$.



$$V = \frac{Qd\cos\theta}{4\pi\epsilon_0 r^2}$$

or in general, for a dipole at any orientation, positioned off-origin:

$$V = \frac{1}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^2} \mathbf{p} \cdot \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|}$$

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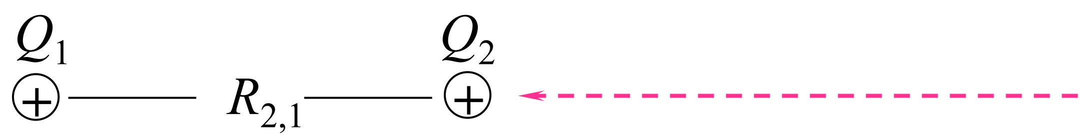
- Potential Energy in a System of Two Point Charges



Q_1 has zero energy if isolated

Charge Q_2 is brought into position from infinity.

The work done in bringing Q_2 into position is:

$$\boxed{W_E(2\ \text{charges}) = Q_2 V_{2,1} = \frac{Q_2 Q_1}{4\pi\epsilon_0 R_{2,1}}}$$

This is the stored energy in the “system”, consisting of the two assembled charges.

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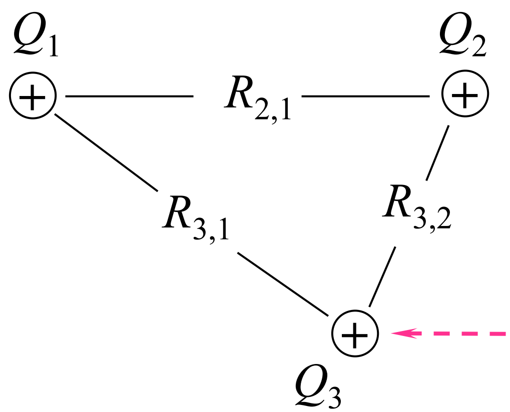
- Potential Energy in a System of Three Point Charges



Charge Q_3 is brought into position from infinity, with Q_1 and Q_2 already situated.

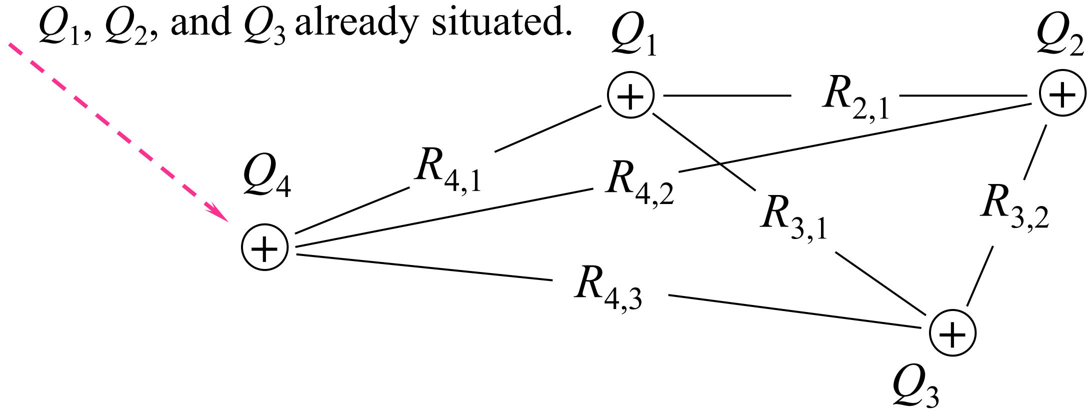
The system energy is now the previous 2-charge energy plus the work done in bringing Q_3 into position:

$$\boxed{W_E(3 \text{ charges}) = Q_2 V_{2,1} + Q_3 V_{3,1} + Q_3 V_{3,2}}$$

Where $V_{3,1} = \dfrac{Q_1}{4\pi\epsilon_0 R_{3,1}}$ and $V_{3,2} = \dfrac{Q_2}{4\pi\epsilon_0 R_{3,2}}$

- Potential Energy in a System of Four Point Charges

Charge Q_4 is brought into position from infinity, with Q_1, Q_2, and Q_3 already situated.



It's getting complicated!

The system energy is now the previous 3-charge energy plus the work done in bringing Q_4 into position:

$$\boxed{W_E(4\ \text{charges}) = Q_2V_{2,1} + Q_3V_{3,1} + Q_3V_{3,2} + Q_4V_{4,1} + Q_4V_{4,2} + Q_4V_{4,3}}$$

where $V_{4,1} = \dfrac{Q_1}{4\pi\epsilon_0 R_{4,1}}$ $\quad V_{4,2} = \dfrac{Q_2}{4\pi\epsilon_0 R_{4,2}}$ and $V_{4,3} = \dfrac{Q_3}{4\pi\epsilon_0 R_{4,3}}$

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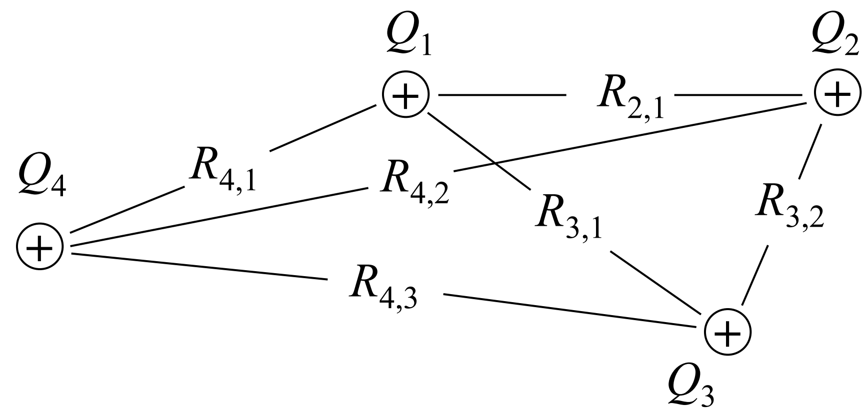
- Potential Energy in a System of Four Point Charges

$$W_E(4\text{ charges}) = Q_2V_{2,1} + Q_3V_{3,1} + Q_3V_{3,2} + Q_4V_{4,1} + Q_4V_{4,2} + Q_4V_{4,3}$$



$$Q_3V_{3,1} = Q_3\frac{Q_1}{4\pi\epsilon_0 R_{13}} = Q_1\frac{Q_3}{4\pi\epsilon_0 R_{31}} = Q_1V_{1,3}$$

$$W_E(4\text{ charges}) = Q_1V_{1,2} + Q_1V_{1,3} + Q_2V_{2,3} + Q_1V_{1,4} + Q_2V_{2,4} + Q_3V_{3,4}$$

$$2W_E = Q_1(V_{1,2}+V_{1,3}+V_{1,4}) + Q_2(V_{2,1}+V_{2,3}+V_{2,4}) + Q_3(V_{3,1}+V_{3,2}+V_{3,4}) + Q_4(V_{4,1}+V_{4,2}+V_{4,3})$$

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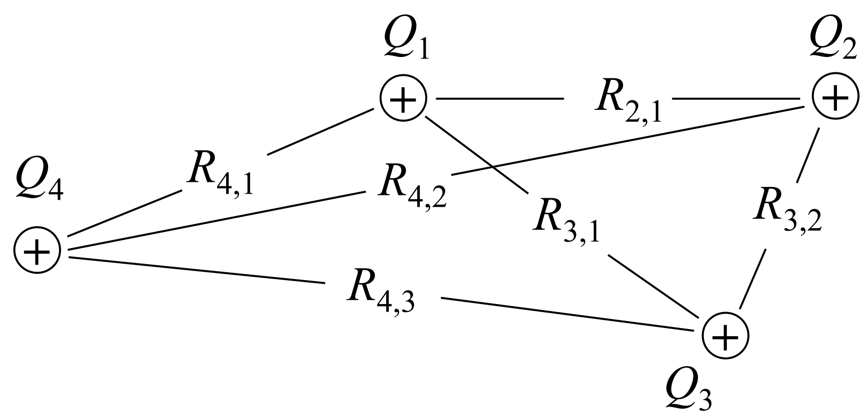
- Condensed Expression for the Stored Energy in an Assembly of Point Charges

$$2W_E = Q_1(V_{1,2}+V_{1,3}+V_{1,4})+Q_2(V_{2,1}+V_{2,3}+V_{2,4})+Q_3(V_{3,1}+V_{3,2}+V_{3,4})+Q_4(V_{4,1}+V_{4,2}+V_{4,3})$$

Define the “local” potentials:

$$\begin{cases} V_1 = V_{1,2}+V_{1,3}+V_{1,4} \\ V_2 = V_{2,1}+V_{2,3}+V_{2,4} \\ V_3 = V_{3,1}+V_{3,2}+V_{3,4} \\ V_4 = V_{4,1}+V_{4,2}+V_{4,3} \end{cases}$$



..so that $W_E(4\text{ charges}) = \frac{1}{2}(Q_1V_1+Q_2V_2+Q_3V_3+Q_4V_4) = \frac{1}{2}\sum_{m=1}^{4} Q_mV_m$

8: Energy and Potential 能量与电位

- Potential Energy in a System: Extension to an *n*-Charge Ensemble

Extending the previous result, we can write the energy expression for *n* charges:

$$\boxed{W_E(n \text{ charges}) = \frac{1}{2}\sum_{m=1}^{n} Q_m V_m}$$

where the local potential
(at the position of charge *m*) is:

$$\boxed{V_m = \sum_{p=1}^{n} V_{m,p} \qquad (p \neq m)}$$

Note that this is the potential due to all charges *except charge m*, evaluated at the location of charge *m*.

• Stored Energy in a Continuous Charge Distribution

If we have a continuous charge, characterized by a charge density function, we use implicitly the expression

$$\frac{1}{2}\sum_{m=1}^{n} Q_m V_m$$

but the charge Q is replaced by the quantity $dq = \rho_v \, dv$, and the summation becomes an integral over the charge volume:

$$\boxed{W_E = \frac{1}{2}\int_{\text{vol}} \rho_v V \, dv}$$

where V is the position-dependent potential function within the charge volume.

- Stored Energy in the Electric Field

$$W_E = \frac{1}{2} \int_{\text{vol}} \rho_v V \, dv$$

We use Maxwell's first equation to express volume charge density in terms of **D**:

$$W_E = \frac{1}{2} \int_{\text{vol}} \rho_v V \, dv = \frac{1}{2} \int_{\text{vol}} (\nabla \cdot \mathbf{D}) V \, dv$$

$$= \frac{1}{2} \int_{\text{vol}} [\nabla \cdot (V\mathbf{D}) - \mathbf{D} \cdot (\nabla V)] \, dv$$

where the vector identity, $\nabla \cdot (V\mathbf{D}) \equiv V(\nabla \cdot \mathbf{D}) + \mathbf{D} \cdot (\nabla V)$, has been used.

Next, the divergence theorem is used on the first term, replacing the volume integral by an integral over the surface that surrounds the volume:

$$W_E = \frac{1}{2} \oint_S (V\mathbf{D}) \cdot d\mathbf{S} - \frac{1}{2} \int_{\text{vol}} \mathbf{D} \cdot (\nabla V) \, dv$$

8: Energy and Potential 能量与电位

- Stored Energy in the Electric Field

We now have: $$W_E = \frac{1}{2}\cancel{\oint_S (V\mathbf{D}) \cdot d\mathbf{S}} - \frac{1}{2}\int_{\text{vol}} \mathbf{D} \cdot (\nabla V)\, dv$$

in which the region of integration now includes *all space*, or wherever the field and potential exist.

At the infinite distance, the potential and **D** fields begin to resemble those of a point charge:

$$V \doteq k_1\left(\frac{1}{r}\right) \qquad D \doteq k_2\left(\frac{1}{r^2}\right) \qquad \text{and therefore:} \quad VD \doteq k_1 k_2\left(\frac{1}{r^3}\right)$$

This means that the surface integral will vanish, because the inverse cube dependence in the integrand falls off at a more rapid rate with r than the surface area increases (surface area increases only as the square of the radius).

- Electric Field Energy and Energy Density

The field energy expression now reads: $$W_E = -\frac{1}{2}\int_{vol} \mathbf{D}\cdot\nabla V\,dv$$

but we know that: $$\mathbf{E} = -\nabla V$$

which leads us to the final result: $$\boxed{W_E = \int_{vol} \frac{1}{2}\mathbf{D}\cdot\mathbf{E}dv}$$

where the *energy density* in the electric field is defined as: $$\boxed{w_E = \frac{1}{2}\mathbf{D}\cdot\mathbf{E} = \frac{1}{2}\epsilon_0 E^2 \quad \mathrm{J/m^3}}$$

HW-4

2. Two point charges, 1 nC at (0, 0, 0.1) and −1 nC at (0, 0, −0.1), are in free space. a) Calculate V at $P(0.3, 0, 0.4)$; b) Calculate $|\mathbf{E}|$ at P; c) Now treat the two charges as a dipole at the origin and find V at P.

4. Three **identical** point charges of 4 pC each are located at the corners of an **equilateral triangle** 0.5 mm on a side in free space. How much work must be done to move one charge to a point **equidistant** from the other two and on the line joining them?

5. Given the electric field $\mathbf{E} = (y+1)\mathbf{a}_x + (x-1)\mathbf{a}_y + 2\mathbf{a}_z$, find the potential difference between the points: a) (2,-2,-1) and (0,0,0); b) (3,2,-1) and (-2,-3,4).

6. Within the cylinder $\rho = 2$, $0 < z < 1$, the potential is given by $V = 100 + 50\rho + 150\rho \sin\varphi$ V. a) Find V, $\mathbf{E}$, $\mathbf{D}$, and ρ_v at $P(1, 60^\circ, 0.5)$ in free space; b) How much charge lies within the cylinder?